Week 7 – Workshop Plan
Fourier Transforms and Potential Fields

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Workshop Overview

This workshop develops intuition for Fourier methods in potential-field geophysics and applies them to gravity and magnetic interpretation. The emphasis is on understanding how sampling, filtering, and phase control what geological information can be recovered.

Learning Objectives

By the end of this workshop, students should be able to:

- Relate spatial wavelength to source depth and geometry
- Recognize aliasing and understand its consequences
- Interpret gravity and magnetic spectra in geological terms
- Apply upward and downward continuation appropriately
- Explain the purpose and limitations of reduction to the pole (RTP)

Timeline and Activities

0–15 min Introduction and framing

- Why Fourier methods appear everywhere in potential-field geophysics
- Review of key ideas: wavelength, sampling, phase
- Overview of the day's activities

15–40 min Demo 1: What a Fourier transform does

- Decomposing a signal into frequency, amplitude, and phase
- Phase as a measure of shift and geometry
- Padding and windowing

Key emphasis: Fourier transforms reorganize information; they do not interpret it.

40–75 min Demo 2: Spatial sampling and aliasing

- Synthetic fold and Basin-and-Range gravity models
- Station spacing and Nyquist wavelength
- Aliasing as coherent but incorrect structure

Key emphasis: Sampling decisions set a hard limit on geological resolution.

75–95 min Real data case study: Nevada gravity

- Complete Bouguer gravity of Nevada
- E–W and N–S power spectra
- Identification of Basin-and-Range wavelengths

95–115 min Isostatic correction and filtering

- Airy isostasy as a physical low-pass filter
- Isostatic residual gravity
- High-pass filtering and wavelength selection

Key emphasis: Filtering encodes geological assumptions.

115–135 min Inversion example

- Parker–Oldenburg sediment thickness inversion
- Interpretation and limitations
- Effect of aliasing on inversion results

135–165 min Magnetic interpretation: RTP and maar geometry

- Magnetic anomalies as vector fields
- Cone/maar geometry and asymmetric anomalies
- Reduction to the pole as a phase operation

Key emphasis: RTP rotates phase; it does not filter noise.

165–180 min Synthesis and discussion

- Gravity vs magnetics: robustness and fragility
- When Fourier methods help—and when they mislead
- Connection to upcoming assignments and field interpretation

Key Takeaways

- Fourier methods are powerful but unforgiving
- Sampling precedes interpretation
- Phase carries geometry
- Geological assumptions are embedded in every filter